

4.1 Develop a program to find the length of the longest palindromic subsequence in a given sequence.

For example:

Input String:

Given sequence is "BBABCBCAB".

Output: 7 (as "BABCBAB" is the longest palindromic subsequence).

"BBBBB" and "BBCBB" are also palindromic subsequences of the given sequence, but not the longest ones.